

THE WATER CYCLE

Simple Learning Material for RAG Testing

This short document is intentionally written in simple language so it can be used to test document upload, chunking, embeddings, pgvector retrieval, and source citations.

Test material: educational synthetic content. It is not an official publication.

1. What is the Water Cycle?

The water cycle is the continuous movement of water between the Earth's surface, the atmosphere, and the ground. Water changes location and sometimes changes its physical state. The same water can move through oceans, clouds, rivers, soil, plants, and living organisms.

Main Stages

Stage	Simple meaning
Evaporation	Liquid water changes into water vapour because of heat.
Condensation	Water vapour cools and changes into tiny liquid droplets.
Precipitation	Water falls from clouds as rain, snow, sleet, or hail.
Collection	Water gathers in oceans, lakes, rivers, soil, and other places.
Infiltration	Some water moves into the soil and can become groundwater.

2. Evaporation and Condensation

Evaporation

Evaporation happens when liquid water receives enough energy to become water vapour. The Sun is the main source of energy that drives evaporation in the natural water cycle. Water can evaporate from seas, lakes, rivers, wet soil, and other exposed surfaces.

Condensation

Condensation happens when water vapour cools. The vapour changes into small liquid droplets. When many droplets collect around tiny particles in the atmosphere, clouds can form.

Easy Example

After a rainy day, a wet road slowly becomes dry. Heat from the surroundings helps some liquid water change into invisible water vapour. This is evaporation.

When water droplets appear on the outside of a cold glass, the water came from water vapour in the air. The vapour cooled and changed into liquid droplets. This is condensation.

3. Precipitation, Collection and Infiltration

Precipitation

Precipitation occurs when water droplets or ice particles in clouds become large enough to fall. Rain is the most common form of precipitation, but snow, sleet, and hail are also forms of precipitation.

Collection

After precipitation reaches the ground, water can collect in rivers, lakes, ponds, wetlands, and oceans. Some water remains on the surface, while some moves through soil.

Infiltration

Infiltration is the movement of water from the ground surface into the soil. Some infiltrated water can continue downward and contribute to groundwater. The rate of infiltration depends on factors such as soil type, vegetation, and how wet the soil already is.

Why the Cycle Matters

The water cycle helps distribute freshwater around the planet. It supports agriculture, ecosystems, drinking-water supplies, and many human activities.

4. Quick Facts

Question	Answer
What provides much of the energy for evaporation?	The Sun.
What process turns water vapour into droplets?	Condensation.
What process describes water falling from clouds?	Precipitation.
What is infiltration?	Movement of water from the ground surface into soil.
Where can collected water be stored?	Oceans, lakes, rivers, ponds, wetlands, and soil.
What is the main idea of the water cycle?	Water continuously moves between Earth's surface, atmosphere, and ground.

Suggested RAG Test Questions

1. What is the water cycle?
2. What is evaporation?
3. Why does condensation happen?
4. What are the forms of precipitation mentioned in the document?
5. What is infiltration?
6. Why is the water cycle important?
7. Which process happens when water vapour cools?
8. What provides much of the energy for evaporation?

Negative test: Ask "What is the capital of Japan?". The document should not provide evidence for that question.